STREAMLINE Testing Data Evaluation Report: 2025-12-05 20:34:31.563518

General Pipeline Settings:

CV Partitions: 3
Partition Method: Stratified
Categorical Cutoff: 10
Statistical Significance Cutoff: 0.05
Random Seed: 42
Run From Notebook: False

Feature Importance/Selection Settings:

Use Mutual Information: True
Use MultiSURF: False
Use TURF: False
TURF Cutoff: 0.5
MultiSURF Instance Subset: 470
Max Features to Keep: 5000
Filter Poor Features: True
Top Features to Display: 40
Export Feature Importance Plot: True

ML Modeling Algorithms:

Artificial Neural Network: False
Category Gradient Boosting: False
Decision Tree: True
Elastic Net: False
ExSTraCS: False
Extreme Gradient Boosting: True
Genetic Programming: False
Gradient Boosting: False
K-Nearest Neighbors: True
Light Gradient Boosting: False
Logistic Regression: True
Naive Bayes: False
Random Forest: True
Support Vector Machine: False
XCS: False
eLCS: False

Modeling Settings:

Primary Metric: balanced_accuracy
Training Subsample for KNN,ANN,SVM,and XGB: 0
Uniform Feature Importance Estimation (Models): True
Hyperparameter Sweep Number of Trials: 100
Hyperparameter Timeout: 900
Export Hyperparameter Sweep Plots: False

LCS Settings (eLCS,XCS,ExSTraCS):

Do LCS Hyperparameter Sweep: False
nu: 1
Training Iterations: 2000
N (Rule Population Size): 2000
LCS Hyperparameter Sweep Timeout: 1200

Stats and Figure Settings:

Export ROC Plot: True
Export PRC Plot: True
Export Metric Boxplots: True
Export Feature Importance Boxplots: True
Metric Weighting Composite FI Plots: balanced_accuracy
Top Model Features To Display: 40

EDA and Processing Settings:

Engineering Missingness Cutoff: 1.0
Cleaning Missingness Cutoff: 1.0
Correlation Removal Threshold: 1.0
List of Exploratory Analysis Ran: ['Univariate Analysis', 'Feature Correlation']
List of Exploratory Plots Saved: []
Use Data Scaling: False
Use Data Imputation: False
Use Multivariate Imputation: False
Overwrite CV Datasets: True

Target Dataset(s):

D1 = SNP_merged_train

Target Data Settings:

Data Path: ./data/SNPData
Output Path: ./Abdoul_Output
Experiment Name: Abdoul_thesis
Class Label: label
Instance Label: sample_id
Match Label: None
Ignored Features: None
Specified Categorical Features: None
Specified Quantitative Features: None

Univariate Analysis of Each Dataset (Top 10 Features for Each)

D1 = SNP_merged_train

Feature	p-value	Test-statistic	Test-name
SNP_17651	2.1508141202396163e-89	401.8757818683847	Chi Square Test
SNP_12955_1	8.953846518973816e-88	394.4367296992009	Chi Square Test
SNP_12955_0	1.6372154316251284e-85	384.0459528695407	Chi Square Test
SNP_2601	3.4710095588269305e-48	212.73833906411048	Chi Square Test
SNP_38243_2	3.499808245379118e-44	194.39046080504087	Chi Square Test
SNP_7709_0	6.292959771420243e-44	193.2229774964071	Chi Square Test
SNP_25744	1.1451334990501064e-43	192.03174941077444	Chi Square Test
SNP_34873	1.3451859121891118e-42	187.13014770723103	Chi Square Test
SNP_33827	4.4517848754257204e-42	184.7492725005196	Chi Square Test
SNP_12094_0	7.260529295851764e-42	183.7762023322693	Chi Square Test

Dataset and Model Prediction Summary: D1 = SNP_merged_train

Data Processing/Counts Summary

	Instances	Total Features	Categorical Features	Quantitative Features	Missing Values	Missing Percent	Class 0	Class 1
Original	470	56717	56717	0	0	0.0	250	220
C1	470	56717	56717	0	0	0.0	220	250
E1	470	56717	56717	0	0	0.0	220	250
C2	470	56717	56717	0	0	0.0	220	250
C3	470	56717	56717	0	0	0.0	220	250
E2	470	67259	67259	0	0	0.0	220	250
C4	470	67259	67259	0	0	0.0	220	250
Processed	470	67259	67259	0	0	0.0	220	250

Cleaning (C) and Engineering (E) Elements

* C1 - Remove instances with no outcome and features to ignore
* E1 - Add missingness features
* C2 - Remove features with invariance or high missingness
* C3 - Remove instances with high missingness
* E2 - Add one-hot-encoding of categorical features
* C4 - Remove highly correlated features

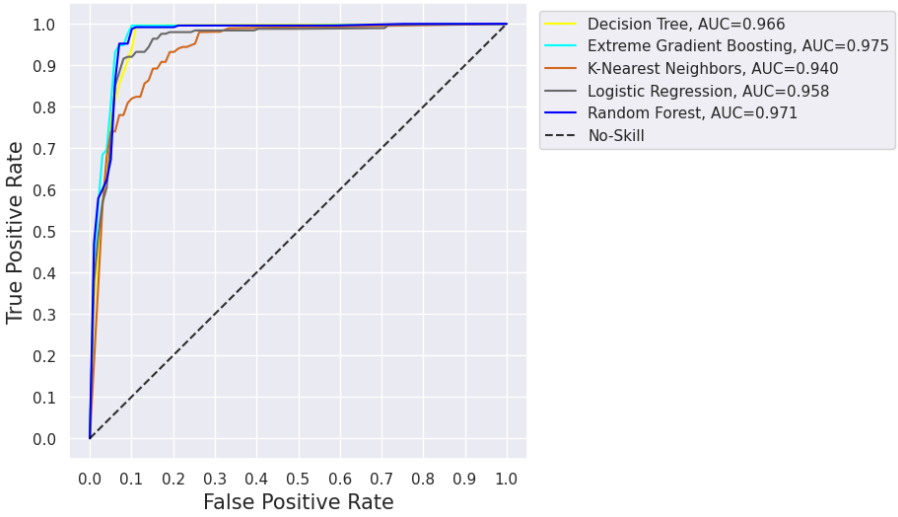
Class Balance (Processed)

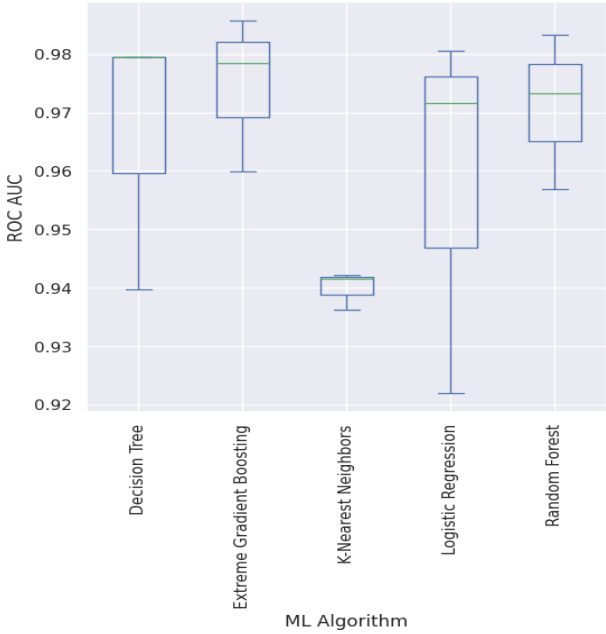
Feature Correlations (Pearson)

No Feature Correlation Plot

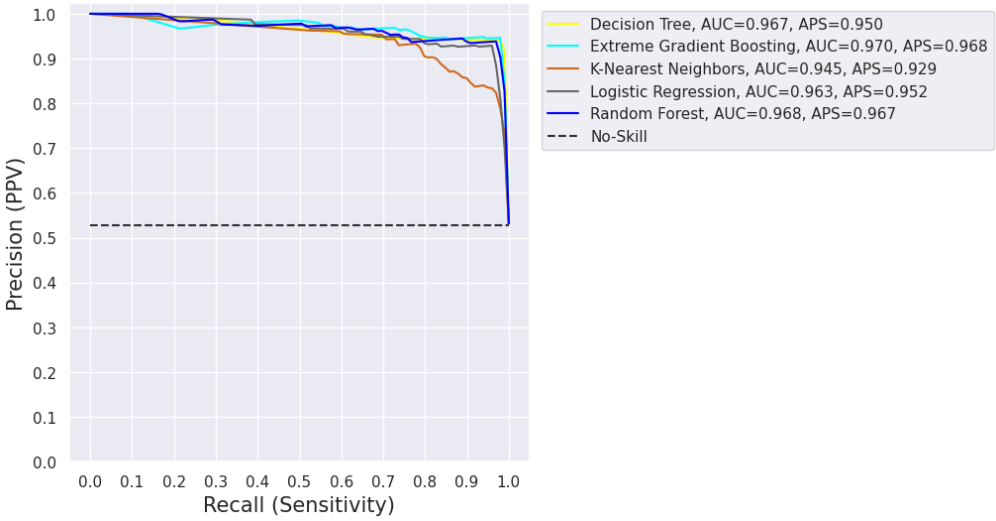
Top ML Algorithm Results (Averaged Over CV Runs):
Best (ROC_AUC): Extreme Gradient Boosting = 0.975
Best (Balanced Acc.): Extreme Gradient Boosting = 0.960
Best (F1 Score): Extreme Gradient Boosting = 0.965
Best (PRC AUC): Extreme Gradient Boosting = 0.970
Best (PRC APS): Extreme Gradient Boosting = 0.968

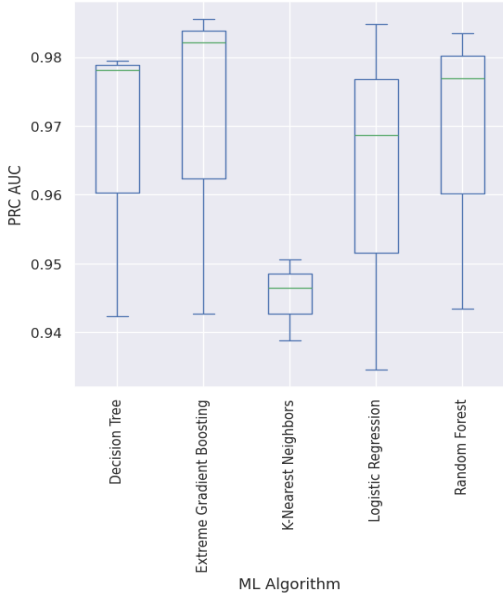
ROC





PRC





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Average Model Prediction Statistics (Rounded to 3 Decimal Points)

D1 = SNP_merged_train

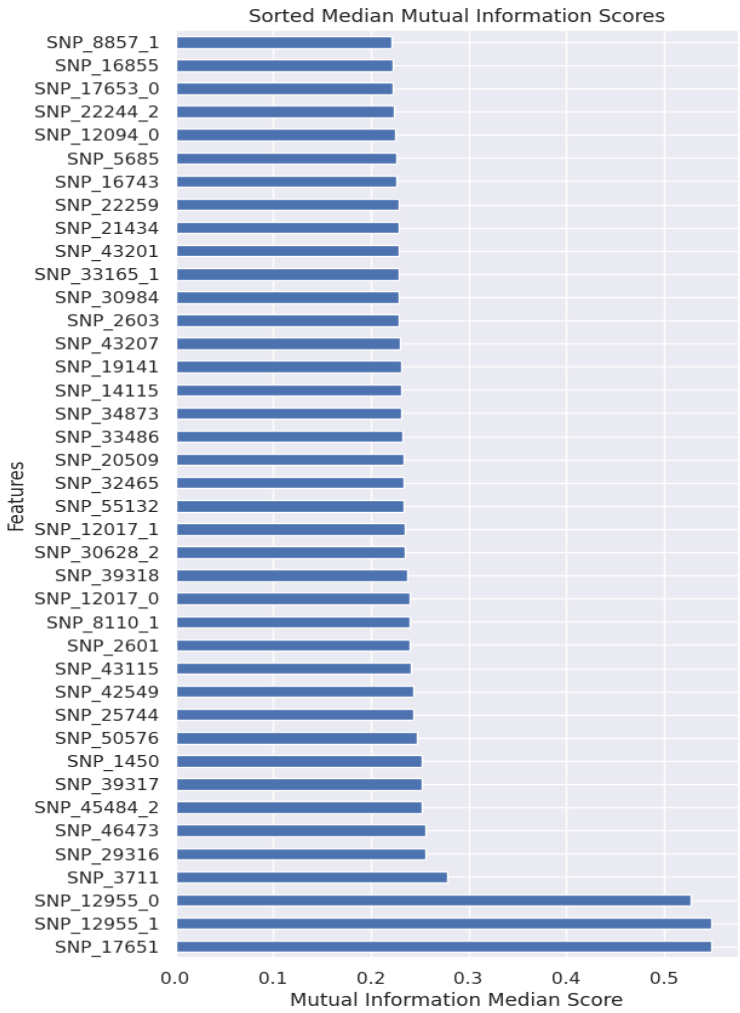
ML Algorithm	Balanced Accuracy	Accuracy	F1 Score	Sensitivity (Recall)	Specificity	Precision (PPV)	TP	TN	FP	FN	NPV	LR+	LR-	ROC AUC	PRC AUC	PRC APS
Decision Tree	0.958	0.96	0.963	0.988	0.927	0.94	82.333	68.0	5.333	1.0	0.985	15.916	0.013	0.966	0.967	0.95
Extreme Gradient Boosting	0.96	0.962	0.965	0.992	0.927	0.94	82.667	68.0	5.333	0.667	0.991	15.891	0.008	0.975	0.97	0.968
K-Nearest Neighbors	0.858	0.857	0.863	0.848	0.868	0.88	70.667	63.667	9.667	12.667	0.836	6.514	0.174	0.94	0.945	0.929
Logistic Regression	0.941	0.943	0.948	0.968	0.913	0.929	80.667	67.0	6.333	2.667	0.961	15.815	0.035	0.958	0.963	0.952
Random Forest	0.948	0.949	0.953	0.968	0.927	0.938	80.667	68.0	5.333	2.667	0.962	14.048	0.035	0.971	0.968	0.967

Median Model Prediction Statistics (Rounded to 3 Decimal Points)

D1 = SNP_merged_train

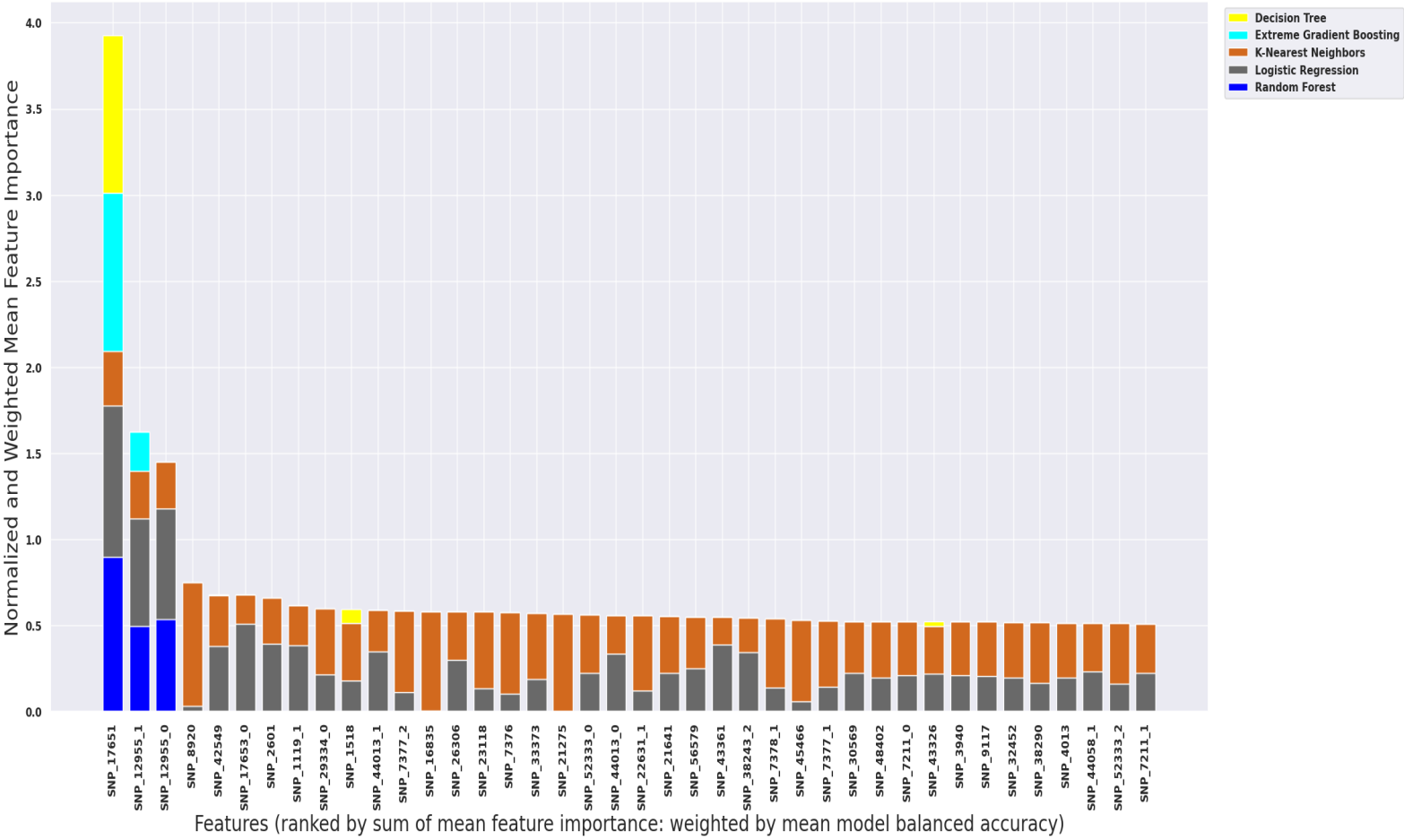
ML Algorithm	Balanced Accuracy	Accuracy	F1 Score	Sensitivity (Recall)	Specificity	Precision (PPV)	TP	TN	FP	FN	NPV	LR+	LR-	ROC AUC	PRC AUC	PRC APS
Decision Tree	0.966	0.968	0.971	0.988	0.932	0.943	82.0	69.0	5.0	1.0	0.986	14.8	0.013	0.98	0.978	0.971
Extreme Gradient Boosting	0.966	0.968	0.97	1.0	0.932	0.943	83.0	69.0	5.0	0.0	1.0	14.8	0.0	0.978	0.982	0.981
K-Nearest Neighbors	0.853	0.854	0.861	0.855	0.863	0.881	71.0	63.0	10.0	12.0	0.84	6.508	0.17	0.942	0.947	0.928
Logistic Regression	0.955	0.955	0.958	0.964	0.946	0.952	81.0	70.0	4.0	3.0	0.959	17.831	0.038	0.972	0.969	0.969
Random Forest	0.955	0.955	0.958	0.964	0.932	0.943	80.0	69.0	5.0	3.0	0.958	14.622	0.038	0.973	0.977	0.977

Feature Importance Summary: D1 = SNP_merged_train



MultiSURF Plot Not Available

Composite Feature Importance Plot (Normalized and Performance Weighted)



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Compare ML Performance Across Datasets

Using Best Performing Algorithms (Kruskall Wallis Compare Datasets)

Pipeline Runtime Summary

SNP_merged_train

Pipeline Component	Phase	Time (sec)
Exploratory Analysis	1	9229.196808576584
Scale and Impute	2	233.5207908153534
Mutual Information (Feature Importance)	3	517.2210321426392
Feature Selection	4	48.57803106307984
Decision Tree(Modeling)	5	55618.45139694214
Extreme Gradient Boosting(Modeling)	5	65773.9516544342
K-Nearest Neighbors(Modeling)	5	114296.28043174744
Logistic Regression(Modeling)	5	54122.1553235054
Random Forest(Modeling)	5	111493.88341975212
Stats Summary	6	58.33893394470215